



UPPSALA
UNIVERSITET

Software Engineering and Project Management

COURSE INFORMATION



UPPSALA
UNIVERSITET

Winter 2024 Team



Jonas Barklund



Marina Shimchenko



Feby Rose



Bilash Deb



UPPSALA
UNIVERSITET

Essential Information

- 5 credits
- Study periods 3 and 4
- Literature:
 - Sommerville, 10th ed.
 - *Software Requirements*, Wiegers & Beatty, 3rd ed.
- Studium is your best friend
- Partial gamified + Roleplaying course



UPPSALA
UNIVERSITET

Course Structure

- **Lectures** (*attendance volunteer*)
 - Theory / class exercises
- **Activities** (*attendance volunteer*)
 - To help fixing concepts / detect misunderstandings
 - Require active participation
- **Group project** (*participation mandatory + 1 meeting/week*)
 - To experience a “real” scenario
 - You must use your free time
 - Group & Individual deliverables



UPPSALA
UNIVERSITET

Let me hear your experience

Why (Software) Projects Fail?



UPPSALA
UNIVERSITET

Course Goals

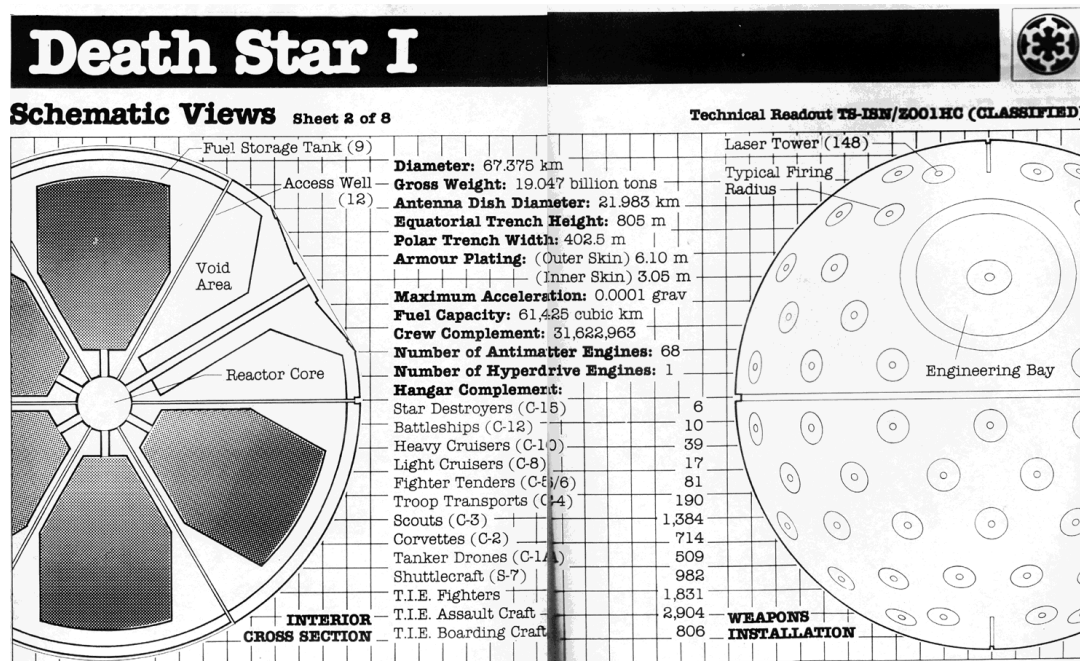
- SEPM is about dealing with large-scale projects (agile is not)





Course Goals

- We cover aspects related with **processes** such as:
 - Requirements definition and specification
 - Prototypes and models
 - Formal specifications

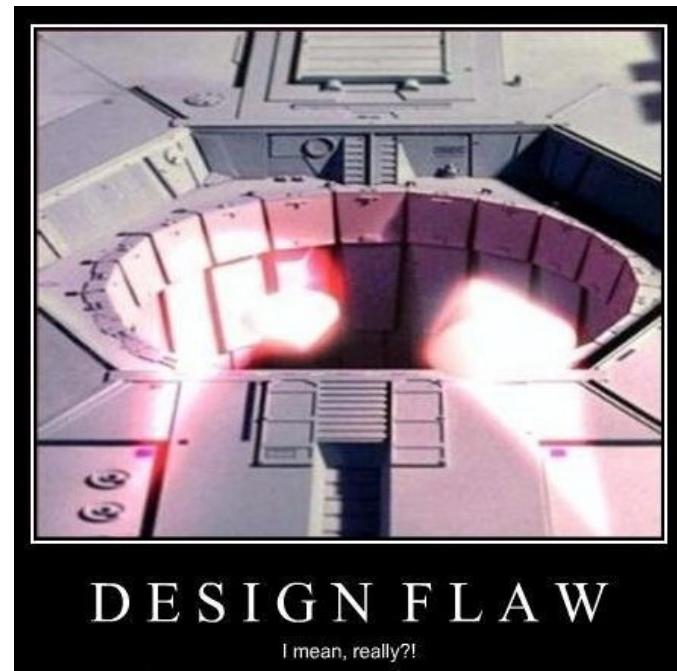




UPPSALA
UNIVERSITET

Course Goals

- We cover aspects related with **quality**:
 - Design methods
 - Maintenance
 - Validation and verification
 - Quality assurance





UPPSALA
UNIVERSITET

Course Goals

- We cover **the human side** of software development:
 - Project planning
 - Project management
 - Human resources





UPPSALA
UNIVERSITET

Course Goals

- SEPM is about people working together in engineering

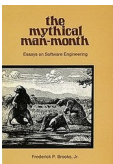
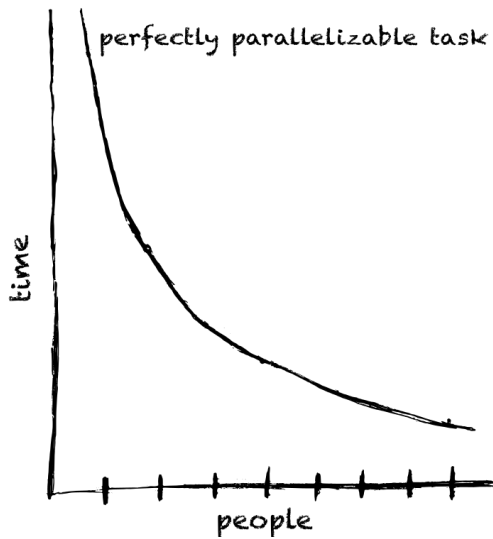




UPPSALA
UNIVERSITET

Course Goals

- SEPM is about people working together in engineering
- Engineering is complex



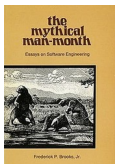
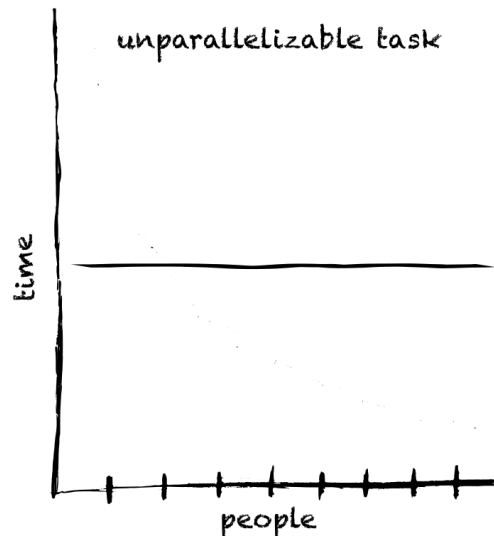
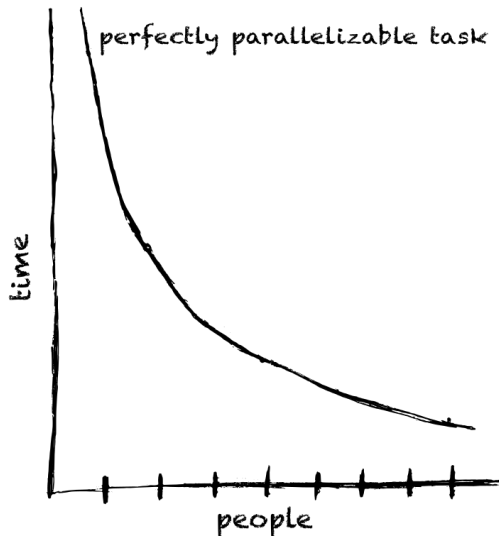
The Mythical Man-Month: Essays on Software Engineering *by Fred Brooks*



UPPSALA
UNIVERSITET

Course Goals

- SEPM is about people working together in engineering
- Engineering is complex



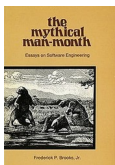
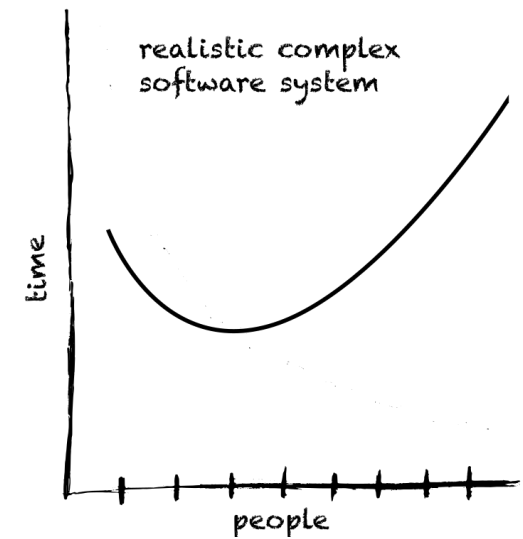
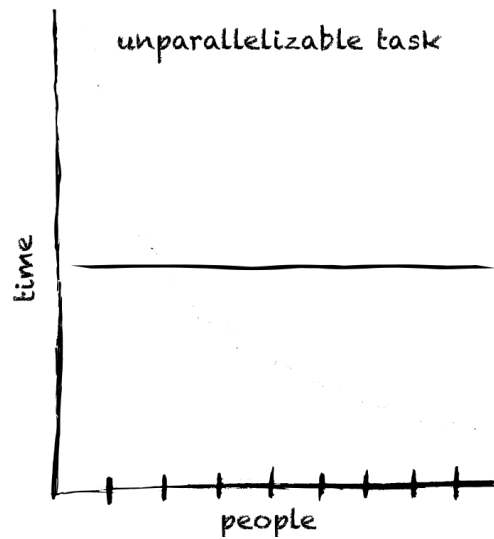
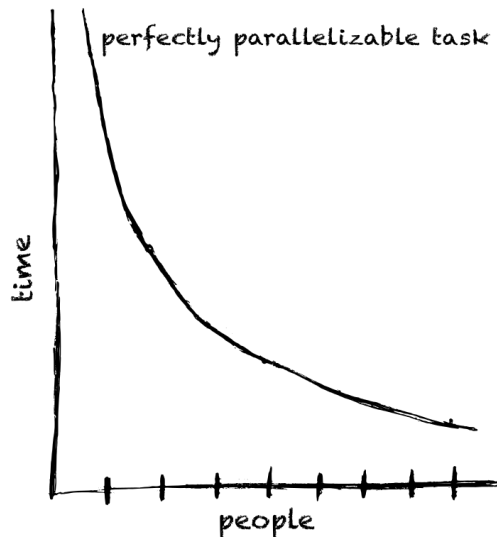
The Mythical Man-Month: Essays on Software Engineering *by Fred Brooks*



UPPSALA
UNIVERSITET

Course Goals

- SEPM is about people working together in engineering
- Engineering is complex



The Mythical Man-Month: Essays on Software Engineering *by Fred Brooks*



UPPSALA
UNIVERSITET

Other SE Courses

Requirements in
Agile Development
(1MD200)

Software Engineering and Project Management (1DL251)

Advanced Software
Design (1DL242)

Software Testing
(1DL610)

Maintenance
Programming
(1DL601)

Complex IT Systems in Large
Organisations (1DL630)

Software Engineering Project (1DL650)



UPPSALA
UNIVERSITET

Course Structure



UPPSALA
UNIVERSITET

Course Structure

- **General information about course**

- Studium:

⋮	▼ Course information (course guidelines, project details, gamification, schedule)	✓ + ⋮
⋮	📄 Course details	✓ ⋮
⋮	📄 Course schedule	✓ ⋮
⋮	📄 Initial questionnaire Sep 1	✓ ⋮

- **General information about Project AND gamification**

- Studium:

⋮	📄 Project Details	✓ ⋮
---	-------------------	-----

- **Weekly schedule (*read in advance*)**

- Studium:

⋮	▼ Week 36 (31.8 - 4.9). Software engineering as a process	Prerequisites: Course information (course guidelines, project details, gamification, schedule)	✓ + ⋮
⋮	📄 Aug. 31 - Software processes		✓ ⋮



UPPSALA
UNIVERSITET

Grading (exam) – 2hp

- Course book (specified chapters)
 - Lectures
 - Activities
-
- The exam has forty (40) points and you need 21
 - Three (3) hours duration.
 - Grade: pass (*Godkänd* - G) / fail (*Underkänd* - U)



UPPSALA
UNIVERSITET

Grading (project) – 3hp

Grade 5	Grade 4	Grade 3	Grade U
Exceptional project quality with extensive delivery of scrum artefacts and design documents, and outstanding team collaboration and presentation.	Functional project quality with complete delivery of scrum artefacts and design documents, and reasonable team collaboration and presentation.	Minimum project quality with delivery of some required scrum artefacts and design documents, and minimum team collaboration and presentation.	Project does not deliver on quality; scrum artefacts and design documents are inappropriate or mostly missing; team functioning and presentation was problematic.
Team has written an excellent D6.	Team has NOT written a D6.	Team has NOT written a D6.	Team has NOT written a D6.
Fully functional software, satisfying the project owner requirements.	Functional software satisfying almost all the project owner requirements.	Functional software with 1 components satisfying almost all the project owner requirements.	Functional software non satisfying the majority of the project owner requirements.
Team has signed and honored at least 1 agreement with another team (See <i>Project overview</i>)	Team has NOT signed or honored at least 1 agreement with another team (See <i>Project overview</i>)	Team has NOT signed or honored at least 1 agreement with another team (See <i>Project overview</i>)	Team has NOT signed or honored at least 1 agreement with another team (See <i>Project overview</i>)



UPPSALA
UNIVERSITET

Our collaboration



Teaching and learning at Uppsala University

Visions for educational activity and development





UPPSALA
UNIVERSITET

Our collaboration

- **Good communication** is a key aspect of (any) the course
 - Please, you are more than welcome to contact me! Anything related!
- **Email communication**
 - I will do my best to answer your emails promptly (put Jonas Barklund in CC)
 - Better if you write through Studium – *it generates an email for you*
 - Please, use a relevant subject line - *[SEPM] Topic*
 - Be complete. If you need help with anything, indicate at least 3 things you tried before writing – *it saves us time and helps me to have a better idea about how to help you*
- **Passing by my office**
 - Not the most effective, better communicate in emails. If we need to meet in person, we can agree to it prior.



UPPSALA
UNIVERSITET

Valuable Attributes

- Academic integrity
- Commitment to learning
- Cooperation & collaboration
- Personal initiative
- Take responsibility
- Punctuality
- Reflective thinking
- Strong work ethic
- **Have fun!**



UPPSALA
UNIVERSITET

Questions

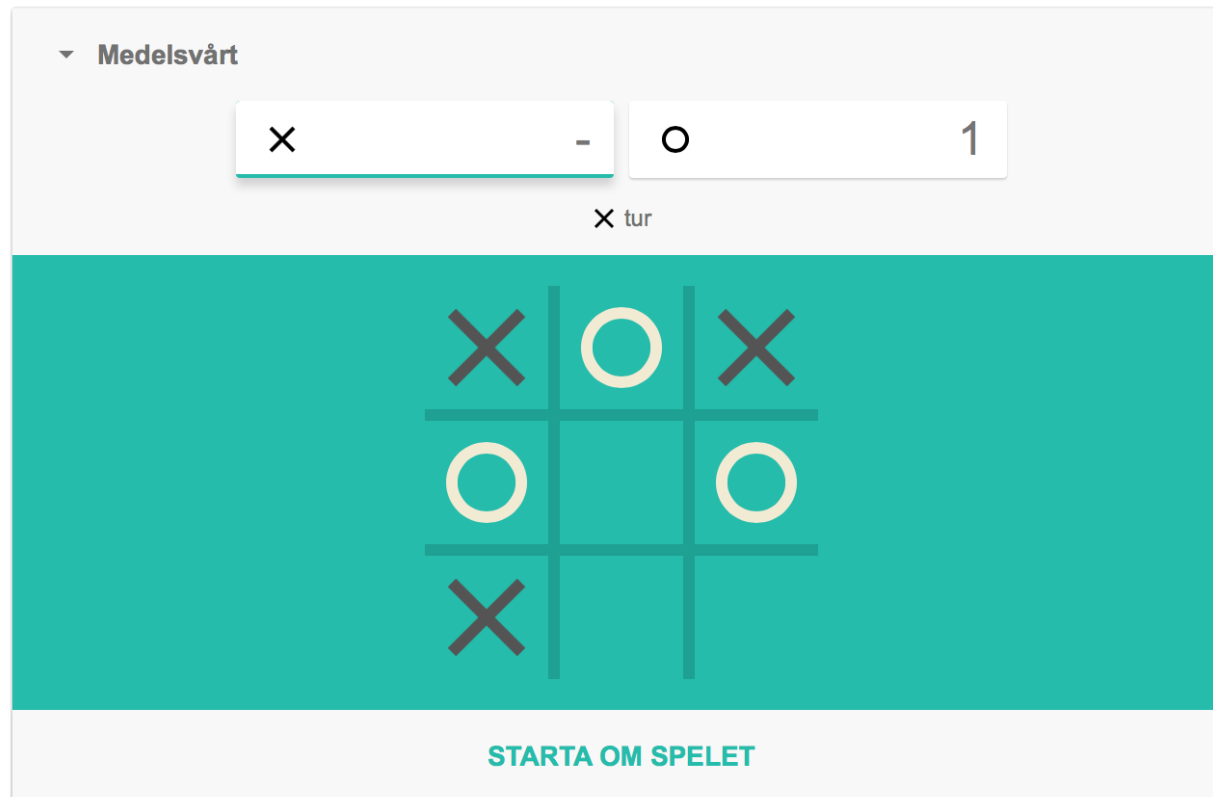
?



UPPSALA
UNIVERSITET

The Impossible Project

- You will build a small 2-players game, like...

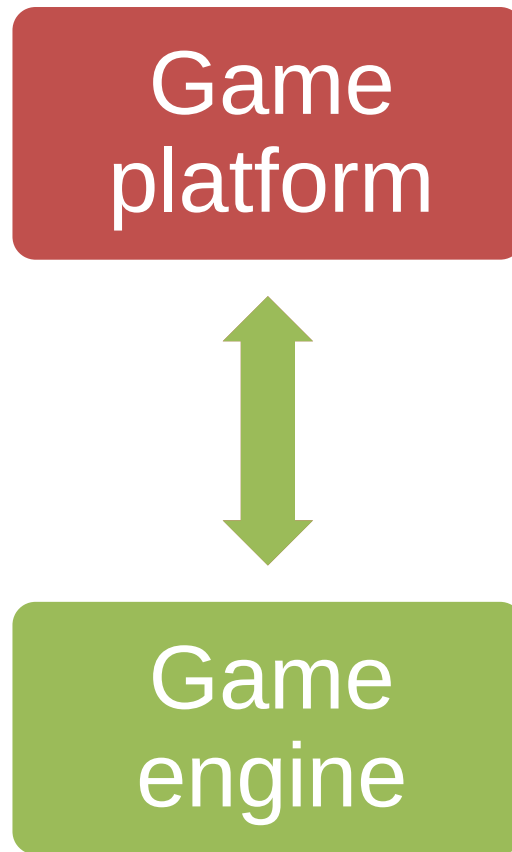




UPPSALA
UNIVERSITET

The Impossible Project

- Main components





UPPSALA
UNIVERSITET

The Impossible Project

- Agile, using Scrum methodology + Kanban boards
- Scrum Coach
 - Retrospective / team evaluation
 - 1 Meeting (1h) *usually* on Fridays every 2 weeks
- Project Owner / Client
 - Initial meeting / Discuss progress



UPPSALA
UNIVERSITET

The ~~im~~possible Project

- **Phase 1**
 - Initial design and development of 1 component
- **Phase 2**
 - Completing the project
 - Proposal:
 - Outsourcing of the remaining component
 - Verification and integration



UPPSALA
UNIVERSITET

The Impossible Project



More details in Project Session 1: 2024-01-19



UPPSALA
UNIVERSITET

Key points!

- **Attendance volunteer except:**
 - Project participation
 - End Sprint meetings with SC
 - Evaluation (written exam + project progress)
- **To read carefully (in Studium):**

☰ ▾ Course information (course guidelines, project details, gamification, schedule)	✓ + ☰
☰ 📄 Course details	✓ ☰
☰ 📄 Course schedule	✓ ☰
☰ ✂ Initial questionnaire Sep 1	✓ ☰
☰ 📄 Project Details	✓ ☰